

1 Methods

1.1 Core Idea

1.1.1 Setup

Consider a logistic regression, with missing features in the observation. Let $x := (x^o, x^m) \in \mathbb{R}^d$. Then the logistic regression is defined as

$$p(y \mid x) := \sigma(w^T x), \quad w \in \mathbb{R}^d.$$

Additionally we have a PC that models the density $p(x)$.

1.1.2 Goal

Compute the prediction in a probabilistically principled manner using

$$p(y \mid x^o) := \mathbb{E}_{p(x^m \mid x^o)} \left[\sigma(w^T x) \right].$$

If it were not for the non-linear sigmoid function σ , we would be able to calculate this in a smooth and decomposable PC, by pulling the Expectation -which is linear- into the leaf nodes. To tackle this we propose to use two tricks.

1.1.3 Trick 1: Pólya-Gamma

To replace the sigmoid function we use the PG-Sigmoid identity, shown in [1], that states:

$$\sigma(z) = \frac{1}{2} \exp\left(\frac{1}{2}z\right) \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(-\frac{1}{2}\gamma z^2\right) \right]$$

for $z \in \mathbb{R}$.

If we set $z := w^T x$, and insert into our expectation, we get:

$$\begin{aligned} p(y \mid x^o) &= \mathbb{E}_{p(x^m \mid x^o)} [\sigma(z)] \\ &= \mathbb{E}_{p(x^m \mid x^o)} \left[\frac{1}{2} \exp\left(\frac{1}{2}z\right) \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(-\frac{1}{2}\gamma z^2\right) \right] \right] \\ &= \frac{1}{2} \mathbb{E}_{p(x^m \mid x^o)} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(\frac{1}{2}z - \frac{1}{2}\gamma z^2\right) \right] \end{aligned}$$

Now, we have a term without a sigmoid, but the x still appears in a quadratic form.

1.1.4 Trick 2: Hubbard-Stratonovich (HS)

To get rid of the quadratic form, we apply the Hubbard–Stratonovich identity [2, 3], that states:

$$\exp\left(-\frac{1}{2}ah^2\right) = \sqrt{\frac{1}{2\pi a}} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2a} - ihu\right) du, \quad (1.1)$$

In plain words, it makes a quadratic term linear at the cost of introducing a helper variable that we need to integrate over.

Currently, in our term, x appears quadratically as well as linearly. To be able to apply HS, we complete the square (again using $z = w^T x$):

$$\begin{aligned} \frac{1}{2}z - \frac{1}{2}\gamma z^2 &= -\frac{1}{2}\gamma \left(z^2 - \frac{z}{\gamma} \right) \\ &= -\frac{1}{2}\gamma \left(\left(z - \frac{1}{2\gamma} \right)^2 - \frac{1}{4\gamma^2} \right) \\ &= -\frac{1}{2}\gamma \left(z - \frac{1}{2\gamma} \right)^2 + \frac{1}{2}\gamma \cdot \frac{1}{4\gamma^2} \\ &= -\frac{1}{2}\gamma \left(z - \frac{1}{2\gamma} \right)^2 + \frac{1}{8\gamma}. \end{aligned} \quad (1.2)$$

Now we apply HS with $a = \gamma$ and $h = z - \frac{1}{2\gamma}$:

$$\begin{aligned}
\exp\left(\frac{1}{2}z - \frac{1}{2}\gamma z^2\right) &= \exp\left(\frac{1}{8\gamma}\right) \exp\left(-\frac{1}{2}\gamma\left(z - \frac{1}{2\gamma}\right)^2\right) \\
&= \exp\left(\frac{1}{8\gamma}\right) \sqrt{\frac{1}{2\pi\gamma}} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2\gamma} - i\left(z - \frac{1}{2\gamma}\right)u\right) du \\
&= \exp\left(\frac{1}{8\gamma}\right) \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi\gamma}} \exp\left(-\frac{u^2}{2\gamma}\right)}_{\mathcal{N}(u;0,\gamma)} \exp\left(-i\left(z - \frac{1}{2\gamma}\right)u\right) du \\
&= \exp\left(\frac{1}{8\gamma}\right) \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(-i\left(z - \frac{1}{2\gamma}\right)u\right) \right] \\
&= \exp\left(\frac{1}{8\gamma}\right) \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(\frac{i u}{2\gamma} - i z u\right) \right] \tag{1.3}
\end{aligned}$$

Substituting back into $p(y \mid x^o)$, we get:

$$\begin{aligned}
p(y \mid x^o) &= \frac{1}{2} \mathbb{E}_{p(x^m|x^o)} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(\frac{1}{2}z - \frac{1}{2}\gamma z^2\right) \right] \\
&= \frac{1}{2} \mathbb{E}_{p(x^m|x^o)} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(\frac{1}{8\gamma}\right) \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(\frac{i u}{2\gamma} - i z u\right) \right] \right] \\
&= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(\frac{1 + 4i u}{8\gamma}\right) \mathbb{E}_{p(x^m|x^o)} [\exp(-i z u)] \right]
\end{aligned}$$

1.1.5 Characteristic Function

The characteristic function of a random variable X is defined as $\phi_X(s) = \mathbb{E}[\exp(isX)]$. The innermost expectation is only over x^m , with x^o fixed, so we split $z = w^T x$ into its observed and missing parts:

$$z = w^T x = w_o^T x^o + w_m^T x^m.$$

The x^o part is constant under $\mathbb{E}_{p(x^m|x^o)}$ and factors out:

$$\begin{aligned}
\mathbb{E}_{p(x^m|x^o)} [\exp(-i z u)] &= \mathbb{E}_{p(x^m|x^o)} \left[\exp\left(-i(w_o^T x^o + w_m^T x^m)u\right) \right] \\
&= \exp\left(-i w_o^T x^o u\right) \cdot \mathbb{E}_{p(x^m|x^o)} \left[\exp\left(-i w_m^T x^m u\right) \right] \\
&= \exp\left(-i w_o^T x^o u\right) \cdot \phi_{x^m|x^o}(-u w_m)
\end{aligned}$$

So the input to the characteristic function is $s := -uw_m$.

1.1.6 Full Formula

Plugging the result of the previous subsection back into the outer expression for $p(y \mid x^o)$, we obtain:

$$\begin{aligned} p(y \mid x^o) &= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp \left(\frac{1 + 4iu}{8\gamma} \right) \cdot \exp \left(-iw_o^T x^o u \right) \cdot \phi_{x^m|x^o}(-uw_m) \right] \\ &= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp \left(\frac{1}{8\gamma} + \frac{iu}{2\gamma} - iw_o^T x^o u \right) \cdot \phi_{x^m|x^o}(-uw_m) \right] \\ &= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp \left(\frac{1}{8\gamma} + iu \left(\frac{1}{2\gamma} - w_o^T x^o \right) \right) \cdot \phi_{x^m|x^o}(-uw_m) \right] \end{aligned}$$

We propose this as a final expression that we can tractably evaluate.

1.2 Implementation

Two ingredients are still abstract in the formula above: the conditional characteristic function $\phi_{x^m|x^o}$, which depends on the density model, and the inner $\mathcal{N}(0, \gamma)$ expectation, which needs to be approximated. This section makes both concrete.

1.2.1 Conditional CF for a GMM

For a Gaussian mixture model $p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \Sigma_k)$ with diagonal covariances $\Sigma_k = \text{diag}(\sigma_k^2)$, the conditional density $p(x^m \mid x^o)$ is again a mixture, with the same component shapes restricted to the missing coordinates and reweighted by the posterior responsibility of each component given x^o :

$$\tilde{\pi}_k(x^o) = \frac{\pi_k \mathcal{N}(x^o; \mu_k^o, \Sigma_k^o)}{\sum_{j=1}^K \pi_j \mathcal{N}(x^o; \mu_j^o, \Sigma_j^o)}.$$

Since the characteristic function of $\mathcal{N}(\mu, \Sigma)$ at s is $\exp \left(is^T \mu - \frac{1}{2} s^T \Sigma s \right)$, and the CF of a mixture is the same convex combination of component CFs, we get a closed-form expression:

$$\phi_{x^m|x^o}(s) = \sum_{k=1}^K \tilde{\pi}_k(x^o) \exp \left(is^T \mu_k^m - \frac{1}{2} s^T \Sigma_k^m s \right),$$

where μ_k^m, Σ_k^m are the component means and covariances restricted to the missing coordinates.

1.2.2 Gauss–Hermite Quadrature

The inner Gaussian expectation is amenable to Gauss–Hermite quadrature, with deterministic nodes $\{t_i\}_{i=1}^N$ and weights $\{\omega_i\}_{i=1}^N$ such that

$$\int_{-\infty}^{\infty} e^{-t^2} g(t) dt \approx \sum_{i=1}^N \omega_i g(t_i)$$

To use this we need to bring the $\mathcal{N}(0, \gamma)$ expectation into the form $\int e^{-t^2} g(t) dt$. Writing the expectation out as an integral against the Gaussian density gives

$$\mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\gamma}} \exp\left(-\frac{u^2}{2\gamma}\right) f(u) du.$$

Apply the substitution $u = \sqrt{2\gamma} t$. It changes three pieces of the integrand:

- the differential: $du = \sqrt{2\gamma} dt$,
- the Gaussian exponent: $\frac{u^2}{2\gamma} = \frac{(\sqrt{2\gamma} t)^2}{2\gamma} = \frac{2\gamma t^2}{2\gamma} = t^2$,
- the argument of f : $f(u) = f(\sqrt{2\gamma} t)$,

and the integration limits are unchanged since $\sqrt{2\gamma} > 0$. Plugging all three substitutions into the integral:

$$\begin{aligned} \mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\gamma}} e^{-t^2} f(\sqrt{2\gamma} t) \sqrt{2\gamma} dt \\ &= \frac{\sqrt{2\gamma}}{\sqrt{2\pi\gamma}} \int_{-\infty}^{\infty} e^{-t^2} f(\sqrt{2\gamma} t) dt. \end{aligned}$$

The constant prefactor simplifies to

$$\frac{\sqrt{2\gamma}}{\sqrt{2\pi\gamma}} = \sqrt{\frac{2\gamma}{2\pi\gamma}} = \sqrt{\frac{1}{\pi}} = \frac{1}{\sqrt{\pi}},$$

so the Gaussian expectation becomes

$$\mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} f(\sqrt{2\gamma} t) dt.$$

Which we then can apply by

$$\mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] \approx \frac{1}{\sqrt{\pi}} \sum_{i=1}^N \omega_i f(\sqrt{2\gamma} t_i).$$

Plugging this into the inner expectation of our final formula, with

$$f(u) = \exp\left(\frac{1}{8\gamma} + iu \left(\frac{1}{2\gamma} - w_o^T x^o\right)\right) \phi_{x^m|x^o}(-uw_m)$$

gives

$$p(y | x^o) \approx \frac{1}{2\sqrt{\pi}} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\sum_{i=1}^N \omega_i \exp\left(\frac{1}{8\gamma} + i\sqrt{2\gamma} t_i \left(\frac{1}{2\gamma} - w_o^T x^o\right)\right) \phi_{x^m|x^o}(-\sqrt{2\gamma} t_i w_m) \right]$$

1.2.3 Approximating the PG integral

We also want don't want to sample the outer expectation over $\gamma \sim \text{PG}(1,0)$ and construct a set of fixed nodes and weights that "approximate" the $\text{PG}(1,0)$ measure best. From what we found, there are multiple ways to do this.

Golub–Welsch construction

One is the *Golub–Welsch algorithm* [4], which produces n -point quadrature nodes and weights from the first $2n$ moments of any positive measure.

The Laplace transform of $\text{PG}(1,0)$ is known in closed form:

$$\mathcal{L}(t) := \mathbb{E}_{\gamma \sim \text{PG}(1,0)}[e^{-t\gamma}] = \text{sech}\left(\sqrt{\frac{t}{2}}\right) = \frac{1}{\cosh(\sqrt{t/2})}.$$

The raw moments follow from derivatives at zero:

$$\mu_k := \mathbb{E}[\gamma^k] = (-1)^k \mathcal{L}^{(k)}(0),$$

which can be evaluated numerically to arbitrary precision.

Given $\mu_0, \dots, \mu_{2n-1}$, the algorithm goes as follows. Form the $n \times n$ Hankel moment matrices

$$H_{ij} = \mu_{i+j}, \quad H'_{ij} = \mu_{i+j+1}, \quad i, j = 0, \dots, n-1.$$

Compute the Cholesky factor $H = R^\top R$ and the Jacobi (tridiagonal) matrix

$$J = R^{-\top} H' R^{-1}, \quad \text{symmetrised as } J \leftarrow \frac{1}{2}(J + J^\top).$$

Eigendecompose $J = V \Lambda V^\top$. Then the quadrature nodes and weights are

$$\gamma_j = \lambda_j, \quad w_j = \mu_0 v_{0j}^2, \quad j = 1, \dots, n,$$

where λ_j are the eigenvalues and v_{0j} is the first component of the j -th eigenvector. The resulting rule integrates polynomials in γ exactly up to degree $2n - 1$.

Nonnegative least-squares construction

The Golub–Welsch rule matches the *moments* of $\text{PG}(1, 0)$, i.e. the Taylor coefficients of $\mathcal{L}$ at $t = 0$, so its accuracy is concentrated near $z = 0$ and decays as $|z|$ grows. An alternative is to match the Laplace transform directly across the range of z .

Fix M candidate support locations $\{\gamma_j\}_{j=1}^M$ log-spaced in $[\gamma_{\min}, \gamma_{\max}]$, and $R \gg M$ fitting points $\{z_r\}_{r=1}^R$ linearly spaced in $[0, z_{\max}]$. Define

$$A_{rj} = e^{-z_r \gamma_j}, \quad b_r = \mathcal{L}(z_r) = \text{sech}\left(\sqrt{z_r/2}\right),$$

and solve the nonnegative least-squares problem

$$\min_{w \geq 0} \|Aw - b\|_2^2.$$

The solution w is then normalized to sum to one, and support points with $w_j < \varepsilon$ are eliminated. The remaining pairs (γ_j, w_j) serve as quadrature nodes and weights for the outer PG integral.

Fully deterministic formula.

Combining PG nodes $(\gamma_j, w_j)_{j=1}^M$ with Gauss–Hermite nodes $(t_i, \omega_i)_{i=1}^N$ yields the deterministic approximation

$$p(y \mid x^o) \approx \frac{1}{2\sqrt{\pi}} \sum_{j=1}^M w_j \sum_{i=1}^N \omega_i \exp\left(\frac{1}{8\gamma_j} + i\sqrt{2\gamma_j} t_i \left(\frac{1}{2\gamma_j} - w_o^T x^o\right)\right) \phi_{x^m \mid x^o}(-\sqrt{2\gamma_j} t_i w_m).$$

1.2.4 Algorithms

Putting the pieces together, we can define three variants of our method, depending on how we want to approximate the integrals. A full Monte Carlo version samples both γ_j and u_i ; a Gauss–Hermite version samples only γ_j and uses quadrature for u_i ; and a fully deterministic version uses quadrature for both.

Full Monte Carlo

Below is an algorithm for the simplest - fully Monte Carlo - variant.

Algorithm 1 PCPG: expected sigmoid under a GMM, full Monte Carlo

Input: LR weights w , partial observation x^o , GMM parameters $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, # PG samples M , # Gaussian samples N

- 1: Split $w = (w_o, w_m)$ and $\mu_k = (\mu_k^o, \mu_k^m)$, $\Sigma_k = (\Sigma_k^o, \Sigma_k^m)$ by the observed/missing mask
- 2: $\tilde{\pi}_k \leftarrow \frac{\pi_k \mathcal{N}(x^o; \mu_k^o, \Sigma_k^o)}{\sum_j \pi_j \mathcal{N}(x^o; \mu_j^o, \Sigma_j^o)}$ for $k = 1, \dots, K$
- 3: $c \leftarrow w_o^T x^o$
- 4: $S \leftarrow 0$
- 5: **for** $j = 1, \dots, M$ **do**
- 6: Sample $\gamma_j \sim \text{PG}(1, 0)$
- 7: **for** $i = 1, \dots, N$ **do**
- 8: Sample $u \sim \mathcal{N}(0, \gamma_j)$
- 9: $s \leftarrow -u w_m$
- 10: $\phi \leftarrow \sum_{k=1}^K \tilde{\pi}_k \exp\left(is^T \mu_k^m - \frac{1}{2}s^T \Sigma_k^m s\right)$
- 11: factor $\leftarrow \exp\left(\frac{1}{8\gamma_j} + iu \left(\frac{1}{2\gamma_j} - c\right)\right)$
- 12: $S \leftarrow S + \text{factor} \cdot \phi$
- 13: **end for**
- 14: **end for**
- 15: **return** $\frac{1}{2MN} \cdot \text{Re}(S)$

Unified version

Since all variant share the same core structure, we don't include separate algorithms for all, but but instead give a unified version that takes the outer and inner nodes/weights as input. Table 1.1 summarises the differences; Algorithm 2 gives the unified procedure.

Variant	Outer (γ_j, w_j)	Inner $(u_i, \omega_i \mid \gamma_j)$	Norm Z
PCPG-MC	M samples PG(1, 0), $w_j=1$	N samples $\mathcal{N}(0, \gamma_j)$, $\omega_i=1$	$\frac{1}{2MN}$
PCPG-GH	M samples PG(1, 0), $w_j=1$	Hermite: $u_i=\sqrt{2\gamma_j} t_i$	$\frac{1}{2\sqrt{\pi}M}$
PCPG-Quad	GW nodes/weights	Hermite: $u_i=\sqrt{2\gamma_j} t_i$	$\frac{1}{2\sqrt{\pi}}$

Table 1.1: The three PCPG variants. GW = Golub–Welsch (§1.2.3); Hermite nodes (t_i, ω_i) from `hermgauss`.

Algorithm 2 PCPG (unified)

Input: LR weights w , partial observation x^o , GMM parameters $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, outer iterates $(\gamma_j, w_j)_{j=1}^M$, inner iterates $(u_i, \omega_i)_{i=1}^N$ (per γ_j), normalisation Z

- 1: Split $w = (w_o, w_m)$ by the observed/missing mask
- 2: $\tilde{\pi}_k \leftarrow \frac{\pi_k \mathcal{N}(x^o; \mu_k^o, \Sigma_k^o)}{\sum_j \pi_j \mathcal{N}(x^o; \mu_j^o, \Sigma_j^o)}$ for $k = 1, \dots, K$
- 3: $c \leftarrow w_o^T x^o$, $S \leftarrow 0$
- 4: **for** $j = 1, \dots, M$ **do**
- 5: **for** $i = 1, \dots, N$ **do**
- 6: $s \leftarrow -u_i w_m$
- 7: $\phi \leftarrow \sum_k \tilde{\pi}_k \exp\left(is^T \mu_k^m - \frac{1}{2}s^T \Sigma_k^m s\right)$
- 8: factor $\leftarrow \exp\left(\frac{1}{8\gamma_j} + iu_i \left(\frac{1}{2\gamma_j} - c\right)\right)$
- 9: $S \leftarrow S + w_j \omega_i \text{factor} \cdot \phi$
- 10: **end for**
- 11: **end for**
- 12: **return** $Z \cdot \text{Re}(S)$

Of course one could also add a forth variant that uses quadrature for the PG integral but sampling for the inner Gaussian, but we don't include it here since it doesn't perform well in practice. The variance of the inner Gaussian sampling is very high which makes the quadrature perform much better.

2 Discussion

2.1 Variance Analysis

2.1.1 Baseline MC Estimator

The MC estimator draws N samples $x^{m(i)} \sim p(x^m \mid x^o)$ and returns

$$\hat{p}_{\text{MC}}(N) = \frac{1}{N} \sum_{i=1}^N \sigma(w^T x^{(i)}).$$

Since $\sigma(\cdot) \in (0, 1)$, the per-sample variance $\sigma_{\text{MC}}^2 := \text{Var}_{p(x^m \mid x^o)}[\sigma(w^T x)]$ is bounded by $1/4$, giving

$$\text{Var}[\hat{p}_{\text{MC}}(N)] = \frac{\sigma_{\text{MC}}^2}{N} \leq \frac{1}{4N}.$$

2.1.2 PCPG-MC Estimator

The PCPG-MC estimator with M outer (PG) and L inner (Gaussian) samples returns

$$\hat{p}_{\text{PCPG}}(M, L) = \frac{1}{2ML} \sum_{j=1}^M \sum_{i=1}^L \text{Re}[h(\gamma_j, u_{ij})],$$

where $h(\gamma, u) := \exp\left(\frac{1}{8\gamma} + iu\left(\frac{1}{2\gamma} - c\right)\right) \cdot \phi_{x^m \mid x^o}(-uw_m)$, $c = w_o^T x^o$, and $u_{ij} \sim \mathcal{N}(0, \gamma_j)$.

Output range. A single sample ($M=L=1$) returns $\frac{1}{2}\text{Re}[h(\gamma, u)]$. Since $|\phi| \leq 1$ and $|h(\gamma, u)| = \exp\left(\frac{1}{8\gamma}\right)$, the value lies in $\left[-\frac{1}{2}e^{1/(8\gamma)}, +\frac{1}{2}e^{1/(8\gamma)}\right]$. Because $\gamma \sim \text{PG}(1, 0)$ can be arbitrarily small, $e^{1/(8\gamma)}$ is unbounded, so individual PCPG samples are not confined to $[0, 1]$ and can be negative or exceed 1. Empirically, with $M=1000$, $L=1$ the standard deviation exceeds 0.76, and with $M=10,000$, $L=1$ it exceeds 1.38. By contrast, every individual MC sample lies in $(0, 1)$.

Variance decomposition. Define the inner conditional expectation $g(\gamma) := \mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[\text{Re } h(\gamma, u)]$ and two variance components:

$$\sigma_B^2 := \text{Var}_{\gamma \sim \text{PG}(1,0)}[g(\gamma)] \quad (\text{outer}), \quad \sigma_A^2 := \mathbb{E}_{\gamma}[\text{Var}_{u|\gamma}[\text{Re } h(\gamma, u)]] \quad (\text{inner}).$$

By the law of total variance:

$$\text{Var}[\hat{p}_{\text{PCPG}}(M, L)] = \frac{1}{4} \left(\frac{\sigma_B^2}{M} + \frac{\sigma_A^2}{ML} \right).$$

These two components behave very differently. σ_B^2 is *finite*: evaluating the Gaussian integral in closed form shows that $g(\gamma)$ remains $O(1)$ as $\gamma \rightarrow 0$ because the $e^{1/(8\gamma)}$ prefactor in h is exactly cancelled by the saddle-point of the Gaussian integral. By contrast, σ_A^2 is *infinite*: for fixed γ , $\text{Var}_{u|\gamma}[\text{Re } h] \approx \frac{1}{2}e^{1/(4\gamma)}$, and the PG density decays only as $\gamma^{-3/4}e^{-1/(8\gamma)}$ near zero, so $\mathbb{E}_{\gamma}[e^{1/(4\gamma)}]$ diverges.

Optimal budget split. Because $\sigma_A^2 = \infty$, the formula above breaks down for $L = 1$: individual samples $\text{Re } h(\gamma, u)$ have unbounded variance due to the unmitigated $e^{1/(8\gamma)}$ factor. The inner average $(1/L) \sum_i \text{Re } h(\gamma, u_i)$ approaches the finite $g(\gamma)$ as L grows, stabilising the estimator. Therefore, for a fixed budget $N = ML$, *more Gaussian inner samples L is better, not more PG samples M* . This is confirmed empirically: at $N=10,000$ the standard deviation falls from 1.38 ($M=10,000, L=1$) to 0.04 ($M=10, L=1,000$). The right strategy is to keep M moderate and let L be large. PCPG-GH takes this to its logical conclusion by making $L \rightarrow \infty$ (exact quadrature) for each PG sample.

2.1.3 PCPG-GH Estimator

Replacing the inner MC with N_H -point Gauss–Hermite quadrature eliminates σ_A^2 entirely. For a fixed number of PG samples M :

$$\text{Var}[\hat{p}_{\text{GH}}(M)] = \frac{\sigma_B^2}{4M}.$$

Compared to PCPG-MC with the optimal split ($L=1$, same total budget $N = M \cdot N_H$ evaluations), PCPG-GH uses $M = N/N_H$ PG samples but removes σ_A^2 :

$$\text{Var}[\hat{p}_{\text{GH}}] = \frac{N_H \sigma_B^2}{4N} \quad \text{vs} \quad \text{Var}[\hat{p}_{\text{PCPG}}] = \frac{\sigma_B^2 + \sigma_A^2}{4N}.$$

PCPG-GH is better when $\sigma_A^2 > (N_H - 1) \sigma_B^2$, i.e. when the inner variance is large relative to the outer. In practice, because the integrand oscillates heavily for small γ

(making σ_A^2 large), PCPG-GH tends to substantially reduce overall variance compared to PCPG-MC. The outer variance σ_B^2/M remains and still exceeds the MC bound σ_{MC}^2/N .

2.1.4 PCPG-Quad Estimator

Replacing both integrals with quadrature (Golub–Welsch for the outer PG integral, Gauss–Hermite for the inner) makes the estimator fully deterministic:

$$\text{Var}[\hat{p}_{\text{Quad}}] = 0.$$

The estimator always returns the same value for the same inputs, and its output is bounded since the PG quadrature nodes γ_j are fixed positive numbers — the factor $e^{1/(8\gamma_j)}$ is a known finite constant for each node. The error is purely from quadrature approximation (bias), not variance.

2.2 Bias Analysis

2.2.1 MC Estimator

MC is unbiased for all $N \geq 1$: sampling from $p(x^m \mid x^o)$ and averaging $\sigma(w^T x)$ is a direct Monte Carlo estimate of the target expectation, so $\mathbb{E}[\hat{p}_{\text{MC}}(N)] = p(y \mid x^o)$ exactly.

2.2.2 PCPG-MC Estimator

PCPG-MC is also unbiased. Both integrals are approximated by ordinary MC, so by linearity of expectation the estimator equals the true double integral, which is $p(y \mid x^o)$ by the PG-sigmoid and Hubbard–Stratonovich identities.

2.2.3 PCPG-GH Estimator

The outer PG integral is still MC (unbiased). The inner integral is replaced by Gauss–Hermite quadrature, which introduces a small approximation error. For N_H quadrature nodes, GH integrates polynomials of degree up to $2N_H - 1$ exactly, and for smooth (analytic) integrands the error decays super-algebraically in N_H . In our setting the integrand is analytic in u , so the bias is negligible already for small N_H (in practice $N_H \geq 20$ is sufficient).

2.2.4 PCPG-Quad Estimator

Both integrals introduce a quadrature error. The Gauss–Hermite inner error is the same as for PCPG-GH. The Golub–Welsch PG quadrature with M nodes integrates polynomials in γ exactly up to degree $2M - 1$; for smooth functions of γ the error again decays rapidly with M . The maximum reliable order is $M \leq 19$ (limited by numerical precision in the moment computation). At $M = 19$ and $N_H \geq 20$, both errors are numerically negligible and the output closely matches the unbiased MC ground truth.

2.3 Comparison

Table 2.1 summarises the properties of all four estimators.

Estimator	Bias	Variance	Output range	Deterministic
MC	none	$\sigma_{\text{MC}}^2/N \leq 1/(4N)$	$(0, 1)$	no
PCPG-MC	none	$(\sigma_B^2/M + \sigma_A^2/ML)/4$	unbounded	no
PCPG-GH	small	$\sigma_B^2/(4M)$	unbounded	no
PCPG-Quad	small	0	fixed	yes

Table 2.1: Summary of estimator properties. $\sigma_{\text{MC}}^2 \leq 1/4$ is bounded; σ_B^2 and σ_A^2 are not.

Variance. MC has the best variance guarantee: $\sigma_{\text{MC}}^2 \leq 1/4$ provides a tight universal bound, independent of the model or weights. The PCPG estimators have no such bound — both σ_B^2 and σ_A^2 depend on the problem and can be large. In the experiments, PCPG-MC with an equal split yields $\sim 26\times$ higher standard deviation than MC at the same budget. PCPG-GH narrows but does not close this gap. PCPG-Quad has zero variance by construction.

Bias. MC and PCPG-MC are exactly unbiased. PCPG-GH and PCPG-Quad introduce small quadrature errors that are negligible at the parameter choices used ($N_H \geq 20$, $M_{\text{PG}} \leq 19$).

When to use which. MC is the simplest and most reliable choice when the density $p(x^m | x^o)$ is easy to sample from and low variance is the priority. PCPG-MC becomes preferable when sampling is difficult but evaluating the characteristic function of $p(x^m | x^o)$ is cheap (as with a GMM), since PCPG never samples the missing features directly. PCPG-GH is a direct improvement over PCPG-MC: it eliminates inner variance at no extra cost for the same outer budget, and should be preferred whenever σ_A^2 is

non-negligible. PCPG-Quad is the right choice when a fully deterministic, reproducible result is required — for example when differentiating through the estimator or when multiple evaluations at the same point must agree exactly.

2.4 Related Work

Here, we just focus on works that also try to calculate

$$\mathbb{E}_{P(x^m|x^o)} \left[\sigma(w^T x) \right]$$

but use a different approach.

2.4.1 Naive Conformant Learning (NaCL)

Khosravi et al. [5] first point out that this expectation is hard to compute in general: even with a naive Bayes feature distribution, it is NP-hard. They also note that the popular shortcut of mean imputation only matches the expectation when the classifier is linear *and* the features are independent — for the sigmoid it is always biased.

Their fix is to pick a feature distribution on which the expectation *is* cheap, namely a naive Bayes model over binary features. The trick is that naive Bayes and logistic regression are dual to each other: every naive Bayes induces a logistic regression, and many naive Bayes models induce the same one. They call a naive Bayes *conformant* with the given LR if both give the same class probabilities on fully observed inputs, and learn such a distribution via geometric programming (“naive conformant learning”). Once this naive Bayes is fitted, the expected prediction is just a marginal inference query and runs in linear time.

Difference to us: they keep the feature distribution simple (naive Bayes on binary inputs) so the expectation is exact and cheap; we keep the feature distribution rich (a GMM or PC on continuous inputs) and instead use a stochastic trick to handle the sigmoid. They also have to refit the feature distribution whenever the classifier changes, while ours is trained once and reused.

2.4.2 Joint Modelling via SAEM

Jiang, Josse, and Lavielle [6] assume a joint Gaussian model on the covariates, $x \sim \mathcal{N}(\mu, \Sigma)$, together with logistic regression on top. Their main contribution is on the training side: they fit (μ, Σ, β) jointly from training data that may itself contain missing values, using a stochastic-approximation EM algorithm with Metropolis–Hastings updates.

For prediction at test time, they simply do plain Monte Carlo: sample $x^{m,(s)} \sim p(x^m \mid x^o)$ from the fitted Gaussian and average,

$$p(y \mid x^o) \approx \frac{1}{S} \sum_{s=1}^S p(y \mid x^o, x^{m,(s)})$$

This is exactly the MC baseline we compare against, just with a single Gaussian instead of a mixture as the density.

Difference to us: they sample the missing features and average the sigmoid; we never sample the missing features, but instead rewrite the sigmoid so that only the characteristic function of $p(x^m \mid x^o)$ is needed. Their feature distribution is also limited to a single Gaussian, while ours can capture multimodal shapes.

Appendix

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